

Program : Diploma in Civil and Environmental Engineering	
Course Code : 4353	Course Title: Sustainable Engineering
Semester : 4	Credits: 3
Course Category: Programme Core course	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To explain the sustainability concepts and atmospheric pollution.
- To illustrate the application of renewable energy resources through case-studies.
- To utilize environmental management standards for assessing impacts of projects.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of Environment	2001	Environmental Science	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain the concepts of sustainability and the means to achieve the same.	10	Understanding
CO2	Explain the effects of air pollution and the means to assess the same.	11	Understanding
CO3	Illustrate the application of renewable energy resources from case-studies.	11	Applying
CO4	Apply the state-of-the-art standards to assess and reduce the environmental impacts of developmental projects.	11	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concepts of sustainability and the means to achieve the same.		
M1.01	Explain the concepts of sustainability	4	Understanding
M1.02	Describe the nexus between Technology and Sustainable development	3	Understanding
M1.03	Explain MDG, SDG and CDM	3	Understanding
Contents: Sustainability: Introduction, concept, evolution of the concept; Social, environmental and economic sustainability concepts; Sustainable development, Nexus between Technology and Sustainable development; Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs), Clean Development Mechanism (CDM).			
CO2	Explain the effects of air pollution and the means to assess the same.		
M2.01	Explain air pollution and its effects.	3	Understanding
M2.02	Describe causes and effects of climate change	3	Understanding
M2.03	Explain the assessment of Climate change	3	Understanding
M2.04	Comprehend the legal provisions for environmental protection	2	Understanding
	Series Test – I	1	

Contents:

Environmental Pollution: Air Pollution and its effects, Water pollution and its sources, Zero waste concept and 3 R concepts in solid waste management; Greenhouse effect, Global warming, Climate change, Ozone layer depletion, Carbon credits, carbon trading and carbon foot print, legal provisions for environmental protection.

CO3	Illustrate the application of renewable energy resources from case-studies.		
M3.01	Explain different types of Energy Resources.	2	Understanding
M3.02	Explain the applications of difference energy resouces	3	Understanding
M3.03	Illustrate relevant case-studies of any one renewable energy resource	6	Applying

Contents:

Resources and its utilisation: Basic concepts of Conventional and non-conventional energy, General idea about solar energy, Fuel cells, Wind energy, Small hydro plants, bio-fuels, Energy derived from oceans and Geothermal energy. Students must be encouraged to refer relevant case-studies of successful application of any one renewable energy resource and discuss/present the same in class.

CO4	Apply the state-of-the-art standards to assess and reduce the environmental impacts of developmental projects.		
M4.01	Explain Environmental Management Standards	3	Understanding
M4.02	Illustrate Life Cycle Analysis of a Project	5	Applying
M4.03	Explain the procedure for Environmental Impact Assessment	3	Understanding
	Series Test -II	1	

Contents:

Environmental management standards: ISO 14001:2015 frame work and benefits, Scope and goal of Life Cycle Analysis (LCA), Circular economy, Bio-mimicking, Environment Impact Assessment (EIA), Industrial ecology and industrial symbiosis.

Text / Reference:

T/R	Book Title/Author
T1	Bradley, A.S; Adebayo,A.O. and Maria, P. Engineering applications in sustainable design and development, Cengage learning
T2	Mackenthun, K.M., Basic Concepts in Environmental Management, Lewis Publication, London

T3	Allen, D. T. and Shonnard, D. R., Sustainability Engineering: Concepts, Design and Case Studies, Prentice Hall.
R1	Environment Impact Assessment Guidelines, Notification of Government of India

Online resources:

Sl.No.	Website Link
1	https://nptel.ac.in/